

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 6: Physical and Chemical Changes

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	6 — Physical and Chemical Changes
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct.

Traps: 'irreversible therefore chemical' (false), 'dramatic/energetic therefore chemical' (false), 'dissolving/state-change makes a new substance' (false), confusing the white ash mass vs ribbon mass, treating crystallisation as chemical, treating galvanising's sacrificial action as mere physical sealing, and reading 'colour change' as automatic proof of chemistry. Do not write on this paper — mark answers on your answer sheet.

1. A teacher writes: 'A change in which one or more new substances are formed.' A student replies that this is also true of dissolving sugar, because sugar-water tastes different. The single best reason the student is wrong is that:

- (A) sugar-water is colourless, and colourless liquids cannot contain new substances
- (B) dissolving always absorbs heat, and only heat-releasing changes can be chemical
- (C) evaporating the water gives back the very same sugar, so no substance with new chemical identity was ever formed
- (D) the sugar particles become too small to see, which proves they were destroyed

2. Equal masses of magnesium ribbon are burnt completely in air; the white ash is weighed. Compared with the original ribbon, the ash:

- (A) weighs less, because burning always destroys part of the metal as smoke
- (B) weighs more, because the magnesium has combined with oxygen from the air to form magnesium oxide
- (C) weighs exactly the same, because mass cannot change in any reaction
- (D) weighs less, because the dazzling light carries away some of the metal's mass

3. Four learners give a 'sure test' for a chemical change. Which test is reliable on its own?

- (A) Check whether the change gave out or took in heat
- (B) Check whether the substance changed colour
- (C) Check whether the change cannot be reversed
- (D) Check whether a substance with new properties has formed that cannot be undone by a simple physical method

4. A student argues: 'Melting wax is reversible, so all reversible changes are physical, and burning wax is irreversible, so all irreversible changes are chemical.' Which single example most directly breaks the second half of this rule?

- (A) Breaking a ceramic cup — irreversible by simple means, yet no new substance forms, so it is physical
- (B) Boiling water into steam — reversible, so it supports the rule
- (C) Cooking an egg — irreversible and chemical, so it supports the rule
- (D) Freezing water into ice — reversible, so it supports the rule

5. Carbon dioxide is bubbled into clear lime water, turning it milky. If excess CO_2 continues to be passed for a long time, the milkiness can disappear again. The milkiness itself is caused by:

- (A) tiny CO_2 bubbles scattering light, a purely physical cloudiness
- (B) the lime water freezing into fine ice crystals
- (C) a white insoluble substance (calcium carbonate) forming as a precipitate, signalling a chemical change
- (D) dust from the air settling on the surface of the liquid

6. A clean iron nail and a clean copper coin are each left in separate beakers of blue copper sulphate solution. After an hour the iron nail carries a reddish-brown coat and the liquid around it looks greener, while the copper coin is unchanged. The best reason is:

- (A) the coin was too clean to react, while the nail was slightly dirty
- (B) copper sulphate reacts with any metal placed in it, so the coin would also change given more time
- (C) iron is more reactive than copper and displaces it, whereas copper cannot displace itself from copper sulphate, so only the nail reacts
- (D) both metals reacted equally, but the change on the copper coin is simply invisible

7. Three identical bare iron nails sit in three tubes: Tube A, boiled water sealed under a layer of oil; Tube B, ordinary tap water open to air; Tube C, a tube of dry air kept with a drying agent. After a week, which nail(s) rust?

- (A) Only the nail in Tube B, because it alone has both air (oxygen) and water present
- (B) All three nails, because iron always rusts in time
- (C) Only the nail in Tube C, because dry air is rich in oxygen
- (D) The nails in Tube A and Tube C, but not Tube B

8. Why does a scratch through the zinc coat of a galvanised iron sheet still leave the iron protected, whereas a scratch through ordinary paint soon lets rust start underneath?

- (A) Zinc heals itself by instantly growing back over the scratch
- (B) Zinc is more reactive than iron, so it corrodes in preference and goes on shielding the exposed iron even where the coat is broken
- (C) Zinc is harder than paint, so scratches simply cannot reach the iron
- (D) The scratch in zinc lets in less air than a scratch in paint

9. Hot saturated copper sulphate solution is left to cool slowly and blue crystals grow. A student labels this 'a chemical change because the colour intensified.' The correct verdict is:

- (A) It is a chemical change, because deepening blue means a new compound formed
- (B) It is a chemical change, because heat was needed to dissolve the solid first
- (C) It is neither physical nor chemical, because only the water moved
- (D) It is a physical change (crystallisation); the same copper sulphate simply comes out of solution as pure crystals

10. Which everyday event looks violent or permanent yet is only a physical change?

- (A) a strip of magnesium flaring with blinding light as it burns
- (B) an inflated balloon bursting with a loud bang, leaving the same rubber in torn pieces
- (C) milk slowly turning into thick sour curd overnight
- (D) a lump of coal glowing red and turning to ash in a stove

11. A lit candle is covered with an inverted glass jar; after a short while the flame dies out. Which conclusion is best supported?

- (A) burning is a chemical change that needs a continuous supply of air (oxygen); once the trapped oxygen is used up, the flame dies
- (B) the candle melted the glass, smothering the flame
- (C) the wax suddenly turned into a physical change the moment it was covered
- (D) the jar cooled the flame below its melting point, a purely physical effect

12. Which list is correctly ordered as physical, chemical, chemical?

- (A) Burning of a candle wick; melting candle wax; milk souring
- (B) Dissolving salt in water; crystallising salt back; baking bread
- (C) Souring of milk; melting wax; rusting of a nail
- (D) Melting candle wax; the candle wick burning; milk souring

13. Riya says, 'Any change that gives out heat must be chemical.' To show she is wrong using a PHYSICAL change, the best demonstration is:

- (A) burning a strip of magnesium and feeling intense heat
- (B) stirring anhydrous solid into water and feeling the beaker warm up, yet recovering the same solid on evaporation

- (C) letting wood burn in a fire and feeling it radiate heat
- (D) watching quicklime react with water to form a new substance with heat

14. When water is added to quicklime (used in whitewashing), the mixture hisses, grows very hot, and a new powdery substance forms that no longer behaves like the original quicklime. This change is:

- (A) physical, because only water was added
- (B) a chemical change, because a new substance with different properties forms while heat is released
- (C) crystallisation of the lime
- (D) merely the lime dissolving, fully recoverable by evaporation

15. The burning of natural cooking gas (LPG) at a kitchen stove is classed as a chemical change mainly because:

- (A) the gas simply changes from a liquid in the cylinder to a gas at the burner
- (B) the blue flame shows the gas is only changing colour
- (C) the change can be reversed by cooling the kitchen
- (D) new substances (carbon dioxide and water vapour) with properties unlike the original gas are formed, and the change cannot be undone by simple physical means

16. An object is found to rust faster than an identical one. Which single difference would speed up rusting the MOST?

- (A) keeping it splashed with salty water in open air
- (B) keeping it in a sealed jar with silica-gel drying packets
- (C) keeping it in air that has had all its moisture removed
- (D) coating it thinly with grease and storing it indoors

17. Which pair correctly matches the change to its type?

- (A) Curdling of milk — physical; melting of ghee — chemical
- (B) Rusting of a chain — physical; cutting of cloth — chemical
- (C) Sublimation of camphor — physical; ripening of a mango — chemical
- (D) Digestion of food — physical; freezing of water — chemical

18. A goldsmith hammers a lump of gold into a thin sheet and then into a ring. Throughout, the metal stays gold. The shaping is:

- (A) a chemical change, because hammering forces the metal to react
- (B) a chemical change, because the gold becomes harder
- (C) crystallisation, because the metal forms a new ordered shape
- (D) a physical change, because only the shape alters while the substance remains gold

19. Blue copper sulphate crystals begin to appear only after a hot saturated solution is allowed to cool. The best reason is that:

- (A) cooling chemically converts the water into crystals
- (B) cooling adds fresh copper sulphate to the solution
- (C) a hot solution can hold more dissolved solid than a cold one, so on cooling the excess separates out as crystals
- (D) crystals form only because the blue colour darkens as it cools

20. Magnesium ribbon is often cleaned with sandpaper before it is burnt. The cleaning step is a physical change, but the burning step is chemical. The KEY difference is that during burning:

- (A) the ribbon merely melts and then re-solidifies as the same magnesium
- (B) the sandpaper marks rub off, exposing fresh shiny metal
- (C) the ribbon is only cut into smaller, brighter pieces
- (D) magnesium combines with oxygen to form a new white substance (magnesium oxide) with different properties

21. A student lists clues of a chemical change: (i) a precipitate forms, (ii) the object is cut into pieces, (iii) a gas is given off, (iv) heat or light is produced. How many are genuine clues of a chemical change?

- | | |
|----------|-----------|
| (A) four | (B) three |
| (C) two | (D) one |

22. Burning a magnesium ribbon cannot be undone by simply cooling the white ash, yet freezing water can be undone by gently warming the ice. The KEY reason for this difference is that:

- (A) the ash is colder than the ribbon, so it stays solid
- (B) cooling always reverses burning but never reverses freezing
- (C) magnesium oxide is just powdered magnesium that needs reshaping
- (D) burning is a chemical change that forms a new substance (magnesium oxide), while freezing is only a physical change of state of the same water

23. In which case does the gas appear because a chemical reaction is taking place, rather than because a dissolved or boiled-off gas is simply escaping?

- (A) marble chips (calcium carbonate) reacting with dilute acid to bubble off a gas
- (B) a warm fizzy drink losing its bubbles as it goes flat
- (C) water being boiled into steam
- (D) a bottle of soda releasing gas the moment its cap is opened

24. A precipitate that forms when two solutions are mixed differs from sand stirred into water and then settling to the bottom. The key difference is that:

- (A) the precipitate and the sand are both unchanged solids that simply sank
- (B) the precipitate is a brand-new insoluble substance made by a chemical reaction, whereas the sand was an existing solid that only settled physically

(C) both form only because the liquid was cooled

(D) the precipitate can always be stirred back into solution, but the sand cannot

25. Galvanising and painting both fight rust, but in DIFFERENT ways. Which statement is correct?

(A) Both protect only by forming a barrier, so a scratch ruins both equally

(B) Paint protects chemically and galvanising protects only as a barrier

(C) Galvanising works by turning the iron itself into zinc

(D) Paint protects only as a barrier, while galvanising's zinc also corrodes in place of the iron, protecting it even at a scratch

26. Burning of petrol in an engine and rusting of an iron gate are sometimes compared. Which statement is correct?

(A) Burning is chemical but rusting is just a physical coating

(B) Both are physical changes because the metal and fuel stay the same substances

(C) Both are chemical changes in which a substance combines with oxygen, but burning is fast while rusting is slow

(D) Rusting is chemical but burning is only a change of state of the fuel

27. A wet shirt dries on a line; later, the same shirt left in a heap with food stains develops a sour smell. The drying and the souring are, respectively:

(A) a physical change (water evaporating) and a chemical change (new smelly substances forming)

(B) both physical changes, because the shirt looks unchanged

(C) both chemical changes, because the shirt's fibres react each time

(D) a chemical change (drying) and a physical change (souring)

28. Freshly cut apple slices turn brown in air, but slices brushed with lemon juice or kept under water stay pale far longer. The browning is slowed because:

(A) browning is a chemical change caused by contact with air; keeping air away (a lemon coating or water) slows that chemistry

(B) lemon juice freezes the apple, a physical change that stops browning

(C) the apple browns because of light, which water and lemon block out

(D) browning is purely physical drying, and wetting simply hides it

29. Which set lists ONLY physical changes?

(A) Burning of wood, melting of wax, dissolving of sugar

(B) Cooking of rice, freezing of water, bending of wire

(C) Rusting of iron, breaking of glass, tearing of cloth

(D) Melting of ice, dissolving of salt, cutting of paper

30. One student dissolves sugar in water; another drops a fizzing vitamin tablet into water, which bubbles vigorously. Which statement is correct?

- (A) both are physical changes, because both substances disappear into the water
- (B) dissolving the sugar is physical (the sugar can be recovered), but the fizzing tablet undergoes a chemical change because a new gas is produced
- (C) both are chemical changes, because both solutions look or taste different
- (D) dissolving the sugar is chemical and the fizzing tablet is physical

31. A blacksmith's helper files a rough iron bar smooth, collecting a pile of fine iron filings. The filing is best described as:

- (A) a chemical change, because the iron is broken down into a new substance
- (B) a physical change, because the filings are still iron and only the size has changed, with no new substance formed
- (C) rusting, because filing exposes fresh iron to the air
- (D) crystallisation, because the dust forms tiny grains

32. Three of the following are physical changes; exactly one is a chemical change. Which is the chemical change?

- (A) chopping a vegetable into pieces
- (B) dissolving honey in warm water
- (C) melting chocolate in the sun
- (D) the digestion of food inside the body

33. When an iron nail reacts in copper sulphate solution, which TWO new substances are formed?

- (A) rust and zinc
- (B) more copper sulphate and pure water
- (C) iron oxide and copper oxide
- (D) copper metal (deposited on the nail) and iron sulphate (dissolved in the solution)

34. If an already-rusty iron railing is simply painted over without first removing the rust, rusting often continues beneath the paint. The best reason is that:

- (A) paint chemically turns the rust back into iron
- (B) the paint adds oxygen that speeds up the rusting
- (C) the loose, flaky rust still traps moisture and air against the iron, so the chemical change goes on beneath the paint
- (D) painting over rust always stops every further change permanently

35. Which kitchen change is chemical, because new substances form rather than just a change of shape or state?

- (A) grating a block of cheese into shreds
- (B) frying an egg until the runny white turns firm and opaque
- (C) melting butter in a warm pan
- (D) freezing leftover curry into a solid block

36. Iodine crystals, when gently warmed, turn into a purple vapour and then re-form as crystals on a cool surface above. This change is classed as physical because:

- (A) the purple colour proves a new substance is being made
- (B) the warming chemically splits the iodine into new substances

- (C) the iodine only changes state (solid → vapour → solid) and stays the same substance, so no new substance forms
- (D) the crystals that re-form are chemically different from the first crystals

37. A rubber band is stretched out and then released, snapping back to its original length. This stretching and releasing is best described as:

- (A) a physical change, because the rubber returns to its original shape and no new substance forms
- (B) a chemical change, because the rubber gets slightly warm when stretched
- (C) crystallisation of the rubber as it relaxes
- (D) an irreversible chemical change

38. Glass is heated until it softens, then blown and shaped into a bottle; on cooling it is still glass. The shaping of the hot glass is:

- (A) a chemical change, because the glass was made very hot
- (B) crystallisation, because the glass hardens as it cools
- (C) a physical change, because the glass is only softened and reshaped, staying the same substance
- (D) rusting, because very hot materials always corrode

39. The setting of milk into curd is classed as a chemical change because:

- (A) new substances form, giving the curd a different taste and texture, and the curd cannot be turned back into milk
- (B) the milk only changes from a thin liquid to a thicker liquid, a mere change of state
- (C) cooling the curd will readily turn it back into milk
- (D) no new substance forms; only the colour of the milk changes

40. A student says, 'A change that releases light and great heat, like a burning sparkler, must be chemical.' Is the reasoning sound?

- (A) Yes, because any release of light or heat by itself always proves a chemical change
- (B) No, because light and heat can never accompany a chemical change
- (C) Yes here, because the sparkler forms new substances; but light and heat alone are not always proof, so the real test is whether a new substance forms
- (D) No, because burning a sparkler is actually a physical change of state

41. Air is squeezed into a bicycle tyre with a pump; letting the valve open again releases the same air. Compressing and then releasing the air is:

- (A) a chemical change, because the air becomes warm when compressed
- (B) a physical change, because only the volume and pressure of the same air change, with no new substance formed
- (C) crystallisation of the air inside the tyre
- (D) rusting of the tyre valve

42. A burning incense stick gives off scented smoke, leaves a little ash, and cannot be turned back into a whole stick. This is:

- (A) a chemical change, because new substances (smoke gases and ash) form and the stick cannot be restored
- (B) a physical change, because the stick only changes shape as it slowly shortens
- (C) crystallisation, because the ash collects in a small heap
- (D) merely a change of state, from solid stick to smoke

43. Chrome plating on a steel cycle handle and a sheet of stainless steel both resist rust. The shared reason they resist rusting is that:

- (A) both turn the underlying iron into pure gold
- (B) both keep a rust-resistant surface (a chromium-containing layer or alloy) between the iron and air and moisture
- (C) both make the iron react faster with oxygen, using it up
- (D) both work only because they are always kept under water

44. The coil of an electric heater glows orange when switched on and returns to dull grey when switched off, staying the same metal throughout. The change in colour is:

- (A) a chemical change, because the coil reacts with the electricity
- (B) a chemical change, because heat always makes new substances
- (C) a physical effect: the hot metal merely glows and returns to the same metal, so no new substance forms
- (D) crystallisation, because the metal coil changes colour

45. Sublimation of camphor, melting of ice, and bending of a metal spoon are grouped together by a student. What do all three share?

- (A) In each, a new substance forms, so all three are chemical changes
- (B) In each, the substance keeps its identity, so all three are physical changes
- (C) Each gives off a gas, so all three are chemical changes
- (D) Each forms a precipitate, so all three are chemical changes

46. Which statement about rust is correct?

- (A) rust is just dried mud stuck onto the iron's surface
- (B) rust is the same iron, only a different colour
- (C) rust is harder and stronger than the original iron
- (D) rust is a new substance (a reddish-brown oxide of iron) formed from iron, oxygen and moisture, and it is flaky and weaker than iron

47. A simple way to slow the rusting of stored steel tools, without applying any coating at all, is to:

- (A) store them in a damp cellar
- (B) sprinkle a little salt water on them each week
- (C) wrap them in a wet cloth
- (D) keep them in a dry place, for example a box with a moisture-absorbing (silica-gel) sachet, so the moisture needed for rusting is kept away

48. Which change can be reversed, giving back the original material?

- (A) burning the wick and wax of a candle
- (B) cooking rice grains until they turn soft
- (C) the souring of milk into curd
- (D) melting solid wax and letting it cool and set into solid wax again

49. A teacher demonstrates that limewater turns milky with CO_2 . A student suggests blowing exhaled breath through a straw into fresh limewater. The expected result and reason are:

- (A) the limewater turns milky, because exhaled breath contains carbon dioxide that reacts to form a white solid
- (B) the limewater turns blue, because breath adds oxygen
- (C) nothing happens, because breath contains only water vapour
- (D) the limewater turns clear and fizzes, because breath is mostly nitrogen

50. Two changes are compared: (i) sugar caramelising in a hot pan to a brown, bitter substance, and (ii) sugar dissolving in cold water. The correct classification is:

- (A) both are physical, because both involve only sugar
- (B) both are chemical, because the sugar changes in each
- (C) caramelising is chemical (a new brown substance forms) and dissolving is physical (sugar is recoverable)
- (D) caramelising is physical and dissolving is chemical

51. A controlled rust experiment must isolate one factor at a time. To test whether MOISTURE is needed (keeping air available), the correct pair of tubes is:

- (A) one nail in open dry air with a drying agent versus one nail in open moist air — both have air, differing only in moisture
- (B) one nail in boiled water under oil versus one in dry air — differing in everything
- (C) one nail in salty water versus one in plain water — both wet
- (D) one nail painted versus one greased — both sealed from air

52. Crystallisation is used to obtain pure, well-shaped crystals of a substance. Which of these is a real-life example of crystallisation?

- (A) obtaining clean sugar-candy (mishri) crystals from a concentrated sugar solution
- (B) burning wood to get a heap of ash
- (C) rusting an iron nail to get a brown powder
- (D) cutting a fruit into neat slices

53. Cooking a raw egg until it sets is irreversible, yet the change of milk into curd is also irreversible. A student concludes 'both are chemical because both are irreversible.' The conclusion happens to be right, but the REASONING is flawed because:

- (A) irreversibility alone does not prove a change is chemical; the real proof is that new substances form, which they do here
- (B) irreversibility always guarantees a chemical change, so the reasoning is perfectly fine

- (C) both changes are in fact physical, so the conclusion is wrong
- (D) neither egg-cooking nor curdling forms a new substance

54. A nail kept in tube A (dry air, drying agent) and a nail in tube B (water that was boiled and sealed under oil) BOTH fail to rust over a week. Together these two tubes show that:

- (A) rusting needs only air, since tube A had air yet did not rust
- (B) rusting needs BOTH air and moisture, since tube A lacks moisture and tube B lacks air, and neither rusts
- (C) rusting needs only moisture, since tube B had water yet did not rust
- (D) iron cannot rust under any laboratory conditions

55. Why can a chemical change usually NOT be reversed by a simple physical method such as cooling or filtering, while many physical changes can?

- (A) because a chemical change forms new substances with new properties, so the originals cannot be restored by a mere physical step, whereas a physical change keeps the same substance
- (B) because physical changes always involve heating and chemical changes always involve cooling
- (C) because chemical changes never give out any visible clue
- (D) because physical changes always form a precipitate

56. Which comparison of dissolving common salt versus burning a candle is CORRECT?

- (A) Both are chemical changes because both make the substance disappear
- (B) Dissolving salt is chemical and burning a candle is physical
- (C) Dissolving salt is physical and reversible by evaporation, while burning a candle is chemical and forms new substances
- (D) Both are physical changes that are easily reversed

57. A change is observed to give out heat, change colour, and form a gas, yet a student insists 'it might still be physical.' What single follow-up test would best settle whether it is chemical?

- (A) Measure exactly how much heat was given out and compare it to a table
- (B) Try to recover the original substances by a simple physical method; if you cannot, a new substance has formed and it is chemical
- (C) Check only whether the colour change was bright enough to see
- (D) Smell the gas and decide based on whether it is pleasant or unpleasant

58. Writing with a piece of chalk on a board leaves white chalk marks; the chalk on the board is still chalk. This is:

- (A) a chemical change, because a new white substance appears on the board
- (B) crystallisation of the chalk onto the board

- (C) a physical change, because the chalk is only ground off onto the board and remains the same chalk
- (D) a chemical change, because the chalk reacts with the surface of the board

59. Which single observation, by itself, would MOST strongly suggest that a chemical change has happened rather than a physical one?

- (A) A solid block is sawn into several smaller blocks
- (B) A shiny metal is bent into a curved shape
- (C) A gas with a completely new smell bubbles out while a colour change appears and the starting material cannot be recovered
- (D) A coloured liquid spreads out and becomes paler when more water is added

60. A student claims: 'Because melting ice and freezing water are reversible, every change of state is reversible and therefore physical.' Which case best supports that changes of state are physical?

- (A) Wax vapour at a candle flame burning to form carbon dioxide and water
- (B) Wood heated until it chars into black charcoal and smoke
- (C) Limestone heated strongly until it gives off a gas and leaves a new solid
- (D) Steam condensing back into liquid water and then re-boiling into steam, the same water throughout

Solutions — Science (Class 7), Chapter 6:

Physical and Chemical Changes — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	B	31	B	41	B	51	A
2	B	12	D	22	D	32	D	42	A	52	A
3	D	13	B	23	A	33	D	43	B	53	A
4	A	14	B	24	B	34	C	44	C	54	B
5	C	15	D	25	D	35	B	45	B	55	A
6	C	16	A	26	C	36	C	46	D	56	C
7	A	17	C	27	A	37	A	47	D	57	B
8	B	18	D	28	A	38	C	48	D	58	C
9	D	19	C	29	D	39	A	49	A	59	C
10	B	20	D	30	B	40	C	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Dissolving is physical: the sugar is still chemically sugar and can be fully recovered by evaporating the water, so no new substance was made — that is the decisive test, not taste. A change is chemical only when a substance of new chemical identity appears. Recovery by a physical means (evaporation) is the proof here. Colourlessness is irrelevant — many solutions of new compounds are colourless. Heat absorption does not decide the issue: some physical changes absorb or release heat. 'Too small to see' is wrong; the particles are merely dispersed, not destroyed.

2. (B) Burning magnesium combines it with oxygen to make magnesium oxide (MgO), so the ash contains the original magnesium PLUS the added oxygen — it is heavier than the ribbon. This is a chemical change (a new substance with new properties). 'Burning destroys metal as smoke' is the common misconception; here the product is a solid that gained oxygen. 'Same mass' confuses the conservation idea (total mass including the air's oxygen is conserved, but the solid alone gains mass). Light does not remove mass.

3. (D) The only dependable signature of a chemical change is the formation of a new substance with new properties, not recoverable by a physical step. Heat exchange happens in many physical changes too (dissolving washing soda warms; melting ice cools). Colour change can be physical (heating a wire glows; some indicators are merely diluted). Irreversibility alone is false — breaking glass is irreversible yet physical. So only the new-substance criterion holds up alone.

4. (A) The rule 'irreversible $\Rightarrow$ chemical' is false, and the cleanest counter-example is a smashed cup: you cannot reassemble it, yet the pieces are still the same material — no new substance, so it is physical. Boiling and freezing are reversible examples that do not test the irreversible-half claim. Cooking an egg is genuinely irreversible AND chemical, so it agrees with the false rule rather than refuting it. Only the broken cup shows irreversible-but-physical.

5. (C) CO₂ reacts with lime water to form insoluble calcium carbonate, whose fine white particles make the liquid look milky — a precipitate is a classic chemical-change clue. The new solid has different properties from the starting materials. Scattering by gas bubbles is not the cause; the cloudiness is a solid product. No freezing occurs at room temperature. Dust is irrelevant — the milkiess appears the instant CO₂ reacts, in a sealed clean tube too.

6. (C) Displacement depends on reactivity: iron is more reactive than copper, so it pushes copper out of copper sulphate, depositing copper on the nail and turning the liquid green (iron sulphate). A copper coin cannot displace copper from copper sulphate — a metal cannot push itself out of its own salt — so the coin is unchanged. Cleanliness is not the deciding factor. Copper sulphate does not react with every metal; only one more reactive than copper will react. The coin's lack of change is real, not merely invisible.

7. (A) Rusting needs iron, oxygen (from air) AND moisture together. Tube A has water but the boiling removed dissolved air and the oil seals air out — no oxygen, so no rust. Tube C has air but no moisture — no rust. Only Tube B supplies both, so only B rusts. 'Iron always rusts' ignores that removing either condition stops it. Dry air alone (C) cannot rust iron because moisture is missing.

8. (B) Galvanising works partly as a barrier but crucially because zinc is more reactive than iron: at a scratch the zinc is attacked first and keeps protecting the bare iron (sacrificial action). Paint is only a barrier — once breached, the exposed iron rusts. Zinc does not magically regrow. Hardness is not why it protects; even a deep scratch is still guarded by the surrounding zinc's chemistry. Air entry is similar for both scratches, so that is not the reason.

9. (D) Crystallisation separates the dissolved copper sulphate as crystals of the very same substance — no new substance forms, so it is physical, used to obtain pure crystals from a solution. The colour deepens only because solid crystals concentrate the blue; no new

compound appears. Heating to dissolve does not make it chemical — that step is also physical (dissolving). It is definitely a change (a state/phase change of the solute), so 'neither' is wrong.

10. (B) A bursting balloon is loud and sudden, but the rubber is merely torn — it is still the same rubber, so no new substance forms and the change is physical. Burning magnesium forms a new substance (magnesium oxide) with light and heat — chemical. Milk souring into curd makes new substances — chemical. Coal glowing and turning to ash is combustion forming new substances — chemical. Only the balloon, despite the bang, keeps the same substance throughout.

11. (A) Burning a candle is a chemical change that needs a steady supply of oxygen from the air. Under the inverted jar, the trapped air holds only a limited amount of oxygen; once that is used up, the flame goes out — showing burning consumes oxygen. The glass is nowhere near hot enough to melt. Covering the flame does not switch the change from chemical to physical. The flame is extinguished by lack of oxygen, not by being cooled below a 'melting point'.

12. (D) Melting wax keeps the same substance (physical); a burning wick makes new substances such as carbon dioxide and water (chemical); souring milk forms new acidic substances (chemical) — so physical, chemical, chemical. The option starting with a burning wick puts chemical first. 'Dissolving, crystallising, baking' is physical, physical, chemical. 'Souring, melting, rusting' is chemical, physical, chemical. Only the first option matches the required order.

13. (B) Some dissolving processes release heat yet form no new substance — the solid can be recovered by evaporation, proving the warm change was physical. That defeats 'heat $\Rightarrow$ chemical'. Burning magnesium and burning wood are chemical (new substances), so they support Riya rather than refute her. Quicklime plus water forms a brand-new substance (slaked lime) with heat — that is chemical, so it cannot serve as the physical counter-example.

14. (B) Adding water to quicklime produces slaked lime — a brand-new substance with properties unlike quicklime — together with a strong release of heat, so it is a chemical change. It is more than 'just adding water': the substance's identity changes. It is not crystallisation, which gives crystals of the same substance from a solution. And it cannot be undone by evaporating, because a new compound has formed; the change is not a simple, recoverable dissolving.

15. (D) Burning LPG combines the fuel with oxygen to make new substances — carbon dioxide and water vapour — whose properties differ from the original gas, and the change cannot be reversed by simple physical means; that is the hallmark of a chemical change. The liquid-to-gas step at the cylinder is a separate physical change, not the reason burning

is chemical. The blue flame is not 'just a colour change'. Cooling the kitchen cannot turn the products back into LPG.

16. (A) Salt water in open air supplies oxygen and abundant moisture and the dissolved salt makes the water conduct better, all of which speed up rusting most. Silica-gel removes moisture, so rusting is slowed, not sped up. Dry air lacks the needed moisture, again slowing rust. Grease seals out air and moisture, the slowest case. So salty water in open air is the fastest-rust condition.

17. (C) Camphor subliming is a state change of the same substance (physical); a mango ripening develops new substances such as sugars and aroma compounds (chemical) — both correct. Curdling is chemical and melting ghee is physical, so that pair is swapped. Rusting is chemical and cutting cloth is physical, also swapped. Digestion is chemical and freezing is physical, again swapped. Only the camphor/mango pair is right.

18. (D) Beating gold changes only its shape; it is still gold, so the change is physical. Hammering does not make gold react with anything — no new substance forms. Any hardness change is a physical property shift, still not a new substance. Crystallisation specifically means crystals coming out of a solution, which has nothing to do with shaping solid metal by hammering.

19. (C) A hot solution can dissolve and hold more solid than a cold one. As the saturated solution cools, it can no longer keep all the copper sulphate dissolved, so the excess separates out as solid crystals — this is why cooling triggers crystal growth. Cooling is a physical change, not a chemical one, so it does not 'convert water into crystals' or create a new compound. No fresh copper sulphate is added. The deepening colour is a consequence of crystals forming, not the cause.

20. (D) Burning bonds magnesium with oxygen to give magnesium oxide — a new white powder that no longer behaves like the shiny metal, so it is chemical. Cleaning, by contrast, only removed a surface layer (physical). The ribbon does not simply melt back into the same magnesium; it is chemically converted. No mere cutting is involved; the substance's identity itself changes.

21. (B) Precipitate formation, gas evolution, and the giving out of heat or light are recognised clues of a chemical change — that is three. Cutting an object into pieces only changes its size and shape (physical), so it is NOT a clue of a chemical change. Hence three, not four. It is more than two and far more than one, because three of the four items are valid chemical-change indicators.

22. (D) Burning magnesium is a chemical change: oxygen combines with the metal to make a new substance, magnesium oxide, so cooling cannot give the ribbon back. Freezing water is only a physical change of state — the same water — so warming the ice easily reverses it. That difference (new substance vs same substance) is the real reason, not the

ash's temperature. Cooling does not reverse burning, and the white powder is not merely powdered magnesium — its chemical identity has changed.

23. (A) Marble chips reacting with dilute acid produce a brand-new gas (carbon dioxide) as a chemical reaction takes place, turning the marble into new substances — true gas evolution from chemistry. A flat fizzy drink and a freshly opened soda bottle merely release CO₂ that was already dissolved in them — no new substance forms, so those are physical. Boiling water gives steam, which is the same water as a gas (physical). Only the marble-and-acid case makes a gas by a chemical change.

24. (B) A precipitate is a brand-new insoluble substance created when two solutions react chemically — it did not exist before mixing. Sand stirred into water is an already-existing solid that merely sinks; nothing new is made, so that settling is purely physical. They are therefore not 'both unchanged solids'. Neither forms because of cooling. And a true precipitate, being a new insoluble product, generally cannot simply be stirred back into solution the way one might re-suspend sand.

25. (D) Paint is purely a barrier — break it and the bare iron rusts. Galvanising adds a barrier of zinc AND, because zinc is more reactive, the zinc corrodes preferentially and shields the iron even at a scratch. So a scratch does not ruin both equally; galvanising still protects. It is paint that is the simple barrier, not the reverse. Galvanising does not transmute iron into zinc; it merely coats it.

26. (C) Both rusting and burning are chemical changes that involve combining with oxygen — burning is rapid (oxidation with heat and light), rusting is slow oxidation. Rusting forms a new oxide, not a mere coating of the same iron. Neither is physical; in both, new substances appear. Burning is not a change of state — petrol vapour reacts to form new gases, so it is chemical, not just evaporation.

27. (A) Drying is water leaving the cloth as vapour — same water, so physical. The sour smell comes from new substances produced as the stains spoil — that is chemical. They are not both physical: souring makes new compounds. They are not both chemical: evaporation forms no new substance. The labels are not reversed; drying is the physical one and souring the chemical one.

28. (A) The browning of cut apple is a chemical change driven by contact with air. Coating the surface with lemon juice or keeping it under water keeps the air away, so the chemistry is slowed and the slices stay pale. Lemon juice does not freeze the apple. The browning is caused by air, not by light, so blocking light is not the point. It is not mere drying — wetting the apple does not hide a stain but actually slows a real chemical change.

29. (D) Melting ice, dissolving salt, and cutting paper each keep the same substance — all physical. The wood-burning set contains burning (chemical), so it fails. The rice set

contains cooking (chemical). The rusting set contains rusting (chemical). Only the first set is entirely physical, with no new substance formed in any of the three.

30. (B) Dissolving sugar is physical: the sugar keeps its identity and can be recovered by evaporating the water. The fizzing tablet is different — it bubbles because a new gas is produced, a sign that a chemical reaction is taking place, so that change is chemical. They are therefore not both physical. Looking or tasting different does not by itself make dissolving chemical. The labels are not reversed: sugar = physical, fizzing tablet = chemical.

31. (B) Filing produces smaller and smaller bits of iron, but each filing is still iron — no new substance forms — so it is a physical change, like cutting or grinding. It is not chemical: the material's identity is unchanged. It is not rusting; brief filing in air does not corrode the metal. It is not crystallisation, which means crystals separating from a solution, not metal being scraped into dust.

32. (D) Digestion breaks food down into new substances the body can absorb, so it is a chemical change. Chopping a vegetable only changes its size and shape (physical). Dissolving honey disperses it in water, recoverable in principle (physical). Melting chocolate is a change of state of the same chocolate (physical). Only digestion forms new substances, so it is the chemical change among the four.

33. (D) In this displacement reaction the more reactive iron pushes copper out of copper sulphate, so two new substances appear: copper metal, which deposits on the nail as a reddish-brown layer, and iron sulphate, which dissolves to give the pale-green solution. Rust and zinc are not involved. No extra copper sulphate or pure water is produced. The products are not oxides (no oxide forms here) — they are copper metal and iron sulphate.

34. (C) Rust is loose and flaky and holds moisture and air against the iron, so painting straight over it traps exactly the conditions rusting needs; the chemical change simply carries on beneath the paint. Paint cannot turn rust back into iron. Paint does not add oxygen to speed rusting. And painting over rust does not stop further change — the rust must first be removed and the bare metal cleaned for paint to protect properly.

35. (B) Frying an egg turns the runny, clear white into a firm, opaque solid — a new substance that cannot be turned back into raw egg, so it is a chemical change. Grating cheese only changes its size and shape (physical). Melting butter is a change of state of the same butter (physical). Freezing curry just turns liquid to solid, the same curry (physical). Only the frying egg forms new substances, making it the chemical change.

36. (C) Iodine sublimates: it goes straight from solid to vapour on warming and re-forms as solid crystals on a cool surface, but it is iodine throughout — no new substance forms — so the change is physical. The purple colour is just the colour of iodine vapour, not proof of a

new substance. Warming does not split the iodine chemically. The crystals that re-form are chemically identical to the original ones; only the state has changed.

37. (A) Stretching a rubber band and letting it spring back changes only its shape; the rubber is the same substance before and after, with no new substance formed, so it is a physical change (and a reversible one). The slight warmth on stretching does not make it chemical — physical changes can involve small heat effects. It is not crystallisation, and nothing about stretching makes it irreversible or chemical.

38. (C) Softening hot glass and reshaping it into a bottle changes only its shape; the material is still glass, so it is a physical change. Being heated does not by itself make a change chemical — no new substance forms here. Hardening on cooling is not crystallisation in the chemical sense (crystals from a solution). And glass does not 'rust'; rusting is corrosion specific to iron, not a result of being hot.

39. (A) Souring milk into curd produces new substances, giving curd a taste and texture quite unlike milk, and the curd cannot be turned back into milk — the marks of a chemical change. It is not a mere change of state from thin to thick liquid; the milk's substances are chemically altered. Cooling does not turn curd back into milk. New substances certainly do form, so 'only the colour changes' is false.

40. (C) The sparkler is chemical because new substances form when its contents burn; light and heat are supporting clues, not the defining test — some physical processes (a glowing filament, dissolving washing soda) also give light or heat. So the correct reasoning rests on the new-substance test. Light/heat alone is not always proof. They certainly CAN accompany chemical changes, so 'never' is wrong. Burning a sparkler is chemical, not a change of state.

41. (B) Pumping air into a tyre only squeezes the same air into a smaller space, raising its pressure; letting it out gives back the same air. No new substance forms, so it is a physical change. The warmth felt on compression is a physical effect and does not make it chemical. Air does not 'crystallise', and the change has nothing to do with the valve rusting.

42. (A) A burning incense stick forms new substances — smoke gases and ash — and cannot be put back together into a whole stick, so it is a chemical change. It is not merely the stick changing shape as it shortens; its very substance is consumed and converted. Ash gathering in a heap is not crystallisation. And the smoke is a new product of burning, not just the solid stick changing state.

43. (B) Chrome plating coats the iron with rust-resistant chromium, and stainless steel is an iron alloy containing chromium that resists corrosion — both keep a protective chromium-bearing surface against air and moisture. Neither turns iron to gold. They do not speed iron's reaction with oxygen — the opposite, they slow corrosion. They protect in

ordinary air, not only underwater (water would in fact promote rusting of unprotected iron).

44. (C) A heater coil glows when hot and returns to dull grey when it cools, but it is the same metal throughout — the glow is just hot metal giving off light, with no new substance formed — so this colour change is a physical effect. The coil does not 'react with electricity' to make a new substance, heat does not always make new substances, and a colour change alone is not crystallisation. The key point is that no new substance forms.

45. (B) Subliming camphor (solid to gas), melting ice (solid to liquid), and bending a spoon (shape change) all leave the substance chemically unchanged — they are all physical. No new substance forms in any, so 'all chemical' is wrong. Only camphor 'gives off a gas' and even that is its own vapour, not a chemical gas; ice and the spoon give off no gas. None forms a precipitate. The shared feature is unchanged identity.

46. (D) Rust is a genuinely new substance — a reddish-brown oxide of iron — made when iron reacts with oxygen and moisture together; being flaky and weak, it does not protect the metal. It is not dried mud stuck on the surface. It is not 'the same iron, only a different colour' — its chemical identity has changed. And it is not harder or stronger than iron; rust weakens iron, which is why rusting is so damaging.

47. (D) Rusting needs moisture, so storing tools in a dry place — for instance in a box with a silica-gel sachet that soaks up moisture — removes one of the two essentials and slows rusting, with no coating needed. A damp cellar, weekly salt water, or a wet cloth would all supply the moisture (and salt) that speed rusting, doing the opposite of protecting the tools.

48. (D) Melting wax and letting it set again just moves the wax between solid and liquid — the same wax throughout — so it is a reversible physical change. Burning a candle's wick and wax makes new substances that cannot be turned back into a candle (chemical, irreversible). Cooking rice changes it into a new, softened food (chemical, irreversible). Souring makes new acidic substances in curd (chemical, irreversible). Only the wax melting can be reversed to give back the original material.

49. (A) Exhaled breath is rich in carbon dioxide, which reacts with limewater to form an insoluble white solid, turning it milky — the same chemical test, demonstrating a new substance. It does not turn blue; oxygen does not produce that reaction. Breath is not only water vapour — its CO_2 is what reacts. The nitrogen in breath does not react with limewater, so 'clear and fizzing' is wrong; the milkiness is the key result.

50. (C) Caramelising heats sugar until it forms new brown, bitter substances — chemical and not reversible. Dissolving merely disperses sugar in water, and evaporating the water gives the sugar back — physical. They are not both physical: caramelising makes a new

substance. They are not both chemical: dissolving makes none. The labels are not reversed; caramelising is the chemical one.

51. (A) To test moisture alone, both tubes must have air and differ only in moisture: a nail in dry air (drying agent) versus one in moist air does exactly that, isolating moisture as the variable. Boiled-water-under-oil versus dry air changes both air and water, so it is not a clean single-factor test. Salty versus plain water both have moisture, so it tests salt, not moisture. Painted versus greased both block air, testing neither factor cleanly.

52. (A) Growing clean sugar-candy (mishri) crystals from a concentrated sugar solution is crystallisation — pure, well-shaped crystals of the same substance separate out of the solution, a physical change. Burning wood to ash forms new substances (chemical). Rusting an iron nail to brown powder forms a new substance (chemical). Cutting a fruit into slices is just a shape change, not crystallisation. Only the sugar-candy case is true crystallisation.

53. (A) Cooking an egg and curdling milk are chemical because each forms new substances — that is the valid reason, not the mere fact of irreversibility. Irreversibility does NOT guarantee chemistry (breaking glass is irreversible yet physical), so the student's reasoning is flawed even though the verdict is correct. Both changes are chemical, not physical. Both DO form new substances, so the last option is false.

54. (B) Tube A has air but no moisture and does not rust; tube B has moisture but no air and does not rust — so removing EITHER factor stops rusting, proving both air and moisture are needed. 'Only air' is wrong because tube A had air yet stayed shiny. 'Only moisture' is wrong because tube B had water yet stayed shiny. Iron certainly does rust when both factors are present (e.g., open tap water), so the last option is false.

55. (A) A chemical change forms new substances with new properties, so getting the original substances back needs more than a simple physical step like cooling or filtering — that is why most chemical changes cannot be reversed physically. A physical change keeps the same substance, so it can often be restored. Heating-vs-cooling is not the dividing line, chemical changes do often show clues, and forming a precipitate is a chemical clue, not a feature of physical changes.

56. (C) Dissolving salt is physical — the salt is recoverable by evaporating the water. Burning a candle is chemical — the wax vapour forms new substances such as carbon dioxide and water. 'Both chemical because they disappear' is wrong: disappearing into solution is physical. The labels are not swapped; dissolving is the physical one. They are not both physical, because burning cannot be reversed and makes new substances.

57. (B) The decisive test is whether the original substances can be got back by a physical means: if they cannot, a new substance has formed and the change is chemical. Measuring heat alone does not decide it, since physical changes can exchange heat. Brightness of a

colour change is not a reliable criterion. Whether a smell is pleasant or not says nothing about whether a new substance formed; only the recovery test settles it.

58. (C) Writing with chalk only rubs tiny bits of chalk onto the board; the marks are still chalk — the same substance — so no new substance forms and the change is physical. A new white substance is not created; it is simply chalk deposited on the surface. It is not crystallisation, and the chalk does not react chemically with the board. Only its position and size change, which is the signature of a physical change.

59. (C) A new-smelling gas plus a colour change plus failure to recover the starting material together strongly indicate a new substance — chemical. Sawing a block only changes its size and shape (physical). Bending metal changes only shape (physical). Diluting a coloured liquid with water makes it paler but forms no new substance, and the original can be recovered by evaporation (physical). So only the first observation points to chemistry.

60. (D) Water cycling between steam and liquid is a pure change of state — the same water all along, reversible, physical — supporting the claim. Wax vapour burning makes new substances (CO_2 and water), a chemical change, not a simple change of state. Wood charring forms new substances (charcoal, smoke gases), chemical. Heating limestone produces a new gas and a new solid, chemical. Only the steam-water case is a reversible state change.